

# TITAN SMALL BUSINESS SERVICES: SECURITY ARCHITECTURE DOCUMENT (SAD)

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## 1. Perimeter Hardening (UFW & SSH)

- **SSH Status:** Hardened /etc/ssh/sshd\_config by setting PermitRootLogin no and PasswordAuthentication no to prevent brute-force attacks
- **Firewall Logic:** Configured UFW to deny all incoming traffic by default, to minimize the attack surface. Only allowing: 22/tcp (SSH) 8080/TCP (tITAN wIKI)

## 2. The Automated Auditor (Python)

- **Script Logic:**

```
'''python import os DC_IP = "10.0.2.2" LOG_FILE = "/var/log/dc_audit.log"
def check_dc(): response = os.system(f"ping -c 4 {DC_IP} > /dev/null
2>&1") status = "DC is UP" if response == 0 else "DC is DOWN" with
open(LOG_FILE, "a") as f: f.write(status + n)
if name == "main": check_dc() ''' * Telemetry Path: /var/log/sys_audit.log
```

## 3. Containerized App (Docker)

- **Network Isolation:** Utilized a dual-network architecture. The 'titan\_wiki' container acts as a gateway on both frontend and backend networks, while the 'titan\_db' is isolated strictly on the internal backend with no exposed ports, creating "air-gap" from the host.
- **Stack Health:**

| NAME       | IMAGE         | STATUS | PORTS                |
|------------|---------------|--------|----------------------|
| titan_wiki | nginx:alpine  | Up     | 0.0.0.0:8080->80/tcp |
| titan_db   | mariadb:10.11 | Up     | (No ports exposed)   |

## 4. Executive Summary

The Hardened Outpost utilizes a multi-layered defense- in-depth strategy, securing the perimeter via SSH lockdowns and a restrictive firewall policy. System availability is continuously monitored by a custom Python auditor that provides persistent telemetry on critical infrastructure status. The application environment is architected for maximum isolation, ensuring that the database remains invisible to external threats through internal container networking.